

Creation Date 24-Nov-2010

Revision Date 28-Jan-2024

Revision Number 3

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	<u>1,2,4-Trichlorobenzene</u>
Cat No. :	A10130
Synonyms	unsym-Trichlorobenzene; Unsymmetrical trichlorobenzene.; 1,2,4-TCB
Index No	602-087-00-6
CAS No	120-82-1
EC No	204-428-0
Molecular Formula	C6 H3 Cl3
REACH registration number	-

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

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CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Acute oral toxicity
Skin Corrosion/Irritation

Category 4 (H302)
Category 2 (H315)

Environmental hazards

Acute aquatic toxicity
Chronic aquatic toxicity

Category 1 (H400)
Category 1 (H410)

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Warning

Hazard Statements

H315 - Causes skin irritation
H410 - Very toxic to aquatic life with long lasting effects
H302 - Harmful if swallowed

Precautionary Statements

P273 - Avoid release to the environment
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water
P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell

2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
1,2,4-Trichlorobenzene	120-82-1	EEC No. 204-428-0	>95	Acute Tox. 4 (H302) Skin Irrit. 2 (H315) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)

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Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	If symptoms persist, call a physician.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.
Self-Protection of the First Aider	No special precautions required.

4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically. Symptoms may be delayed.
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SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Water spray, carbon dioxide (CO₂), dry chemical, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

No information available.

5.2. Special hazards arising from the substance or mixture

Do not allow run-off from fire-fighting to enter drains or water courses.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂), Phosgene, Hydrogen chloride gas.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

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Ensure adequate ventilation. Use personal protective equipment as required.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Storage Class/LGK 10

Switzerland - Storage of hazardous substances

Storage class - SC 10/12
<https://www.kvu.ch/de/themen/stoffe-und-produkte>
<https://www.kvu.ch/fr/themes/substances-et-produits>
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority.

Component	European Union	The United Kingdom	France	Belgium	Spain
1,2,4-Trichlorobenzene	TWA: 2 ppm (8h) TWA: 15.1 mg/m ³ (8h) STEL: 5 ppm (15min) STEL: 37.8 mg/m ³ (15min) Skin	STEL: 5 ppm 15 min TWA: 1 ppm 8 hr Skin	TWA / VME: 2 ppm (8 heures). restrictive limit TWA / VME: 15.1 mg/m ³ (8 heures). restrictive limit STEL / VLCT: 5 ppm. restrictive limit STEL / VLCT: 37.8	TWA: 2 ppm 8 uren TWA: 15.1 mg/m ³ 8 uren STEL: 5 ppm 15 minuten STEL: 37.8 mg/m ³ 15 minuten Huid	STEL / VLA-EC: 5 ppm (15 minutos). STEL / VLA-EC: 38 mg/m ³ (15 minutos). TWA / VLA-ED: 2 ppm (8 horas) TWA / VLA-ED: 15 mg/m ³ (8 horas)

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			mg/m³, restrictive limit Peau		Piel
Component	Italy	Germany	Portugal	The Netherlands	Finland
1,2,4-Trichlorobenzene	TWA: 2 ppm 8 ore. Time Weighted Average TWA: 15.1 mg/m³ 8 ore. Time Weighted Average STEL: 5 ppm 15 minuti. Short-term STEL: 37.8 mg/m³ 15 minuti. Short-term Pelle	TWA: 0.5 ppm (8 Stunden). AGW - exposure factor 4 TWA: 3.8 mg/m³ (8 Stunden). AGW - exposure factor 4 TWA: 0.38 mg/m³ (8 Stunden). MAK TWA: 0.5 ppm (8 Stunden). MAK Höhepunkt: 0.76 mg/m³ Höhepunkt: 1 ppm Haut	STEL: 5 ppm 15 minutos STEL: 37.8 mg/m³ 15 minutos Ceiling: 5 ppm TWA: 2 ppm 8 horas TWA: 15.1 mg/m³ 8 horas Pele	huid STEL: 37.8 mg/m³ 15 minuten TWA: 7.55 mg/m³ 8 uren	TWA: 2 ppm 8 tunteina TWA: 15 mg/m³ 8 tunteina STEL: 5 ppm 15 minuutteina STEL: 38 mg/m³ 15 minuutteina Iho
Component	Austria	Denmark	Switzerland	Poland	Norway
1,2,4-Trichlorobenzene	Haut MAK-KZGW: 5 ppm 15 Minuten MAK-KZGW: 37.8 mg/m³ 15 Minuten MAK-TMW: 2 ppm 8 Stunden MAK-TMW: 15.1 mg/m³ 8 Stunden	TWA: 2 ppm 8 timer TWA: 15 mg/m³ 8 timer STEL: 38 mg/m³ 15 minutter STEL: 5 ppm 15 minutter Hud		STEL: 30 mg/m³ 15 minutach TWA: 15 mg/m³ 8 godzinach	TWA: 2 ppm 8 timer TWA: 15 mg/m³ 8 timer STEL: 4 ppm 15 minutter. value calculated STEL: 22.5 mg/m³ 15 minutter. value calculated Hud
Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
1,2,4-Trichlorobenzene	TWA: 2 ppm TWA: 15.1 mg/m³ STEL : 5 ppm STEL : 37.8 mg/m³ Skin notation	kože TWA-GVI: 2 ppm 8 satima. TWA-GVI: 15.1 mg/m³ 8 satima. STEL-KGVI: 5 ppm 15 minutama. STEL-KGVI: 37.8 mg/m³ 15 minutama.	TWA: 2 ppm 8 hr. TWA: 15.1 mg/m³ 8 hr. STEL: 5 ppm 15 min STEL: 37.8 mg/m³ 15 min Skin	Skin-potential for cutaneous absorption STEL: 5 ppm STEL: 37.8 mg/m³ TWA: 2 ppm TWA: 15.1 mg/m³	TWA: 15 mg/m³ 8 hodinách. Potential for cutaneous absorption Ceiling: 35 mg/m³
Component	Estonia	Gibraltar	Greece	Hungary	Iceland
1,2,4-Trichlorobenzene	Nahk TWA: 2 ppm 8 tundides. TWA: 15.1 mg/m³ 8 tundides. STEL: 5 ppm 15 minutites. STEL: 37.8 mg/m³ 15 minutites.	Skin notation TWA: 2 ppm 8 hr TWA: 15.1 mg/m³ 8 hr STEL: 5 ppm 15 min STEL: 37.8 mg/m³ 15 min	skin - potential for cutaneous absorption STEL: 5 ppm STEL: 40 mg/m³ TWA: 5 ppm TWA: 40 mg/m³	STEL: 37.8 mg/m³ 15 percekben. CK TWA: 15.1 mg/m³ 8 órában. AK lehetséges borón keresztül felszívódás	STEL: 5 ppm STEL: 37.8 mg/m³ TWA: 2 ppm 8 klukkustundum. TWA: 15 mg/m³ 8 klukkustundum. Skin notation
Component	Latvia	Lithuania	Luxembourg	Malta	Romania
1,2,4-Trichlorobenzene	skin - potential for cutaneous exposure STEL: 5 ppm STEL: 37.8 mg/m³ TWA: 2 ppm TWA: 15.1 mg/m³	TWA: 2 ppm IPRD TWA: 15.1 mg/m³ IPRD Oda STEL: 5 ppm STEL: 37.8 mg/m³	Possibility of significant uptake through the skin TWA: 2 ppm 8 Stunden TWA: 15.1 mg/m³ 8 Stunden STEL: 5 ppm 15 Minuten STEL: 37.8 mg/m³ 15 Minuten	possibility of significant uptake through the skin TWA: 2 ppm TWA: 15.1 mg/m³ STEL: 5 ppm 15 minuti STEL: 37.8 mg/m³ 15 minuti	Skin notation TWA: 2 ppm 8 ore TWA: 15.1 mg/m³ 8 ore STEL: 5 ppm 15 minute STEL: 37.8 mg/m³ 15 minute
Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
1,2,4-Trichlorobenzene		Ceiling: 38 mg/m³ Potential for cutaneous absorption TWA: 2 ppm TWA: 15 mg/m³	TWA: 2 ppm 8 urah TWA: 15.1 mg/m³ 8 urah Koža STEL: 5 ppm 15 minutah STEL: 37.8 mg/m³ 15 minutah	Binding STEL: 5 ppm 15 minuter Binding STEL: 38 mg/m³ 15 minuter TLV: 2 ppm 8 timmar. NGV TLV: 15 mg/m³ 8 timmar. NGV Hud	Deri TWA: 2 ppm 8 saat TWA: 15.1 mg/m³ 8 saat STEL: 5 ppm 15 dakika STEL: 37.8 mg/m³ 15 dakika

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Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

No information available

Predicted No Effect Concentration (PNEC)

No information available.

8.2. Exposure controls

Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Viton (R)	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection

No protective equipment is needed under normal use conditions.

Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

Small scale/Laboratory use

Maintain adequate ventilation

Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

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9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Clear	
Odor	aromatic	
Odor Threshold	No data available	
Melting Point/Range	16 °C / 60.8 °F	
Softening Point	No data available	
Boiling Point/Range	214 °C / 417.2 °F	@ 760 mmHg
Flammability (liquid)	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	Lower 2.5 Upper 6.6	
Flash Point	110 °C / 230 °F	Method - No information available
Autoignition Temperature	571 °C / 1059.8 °F	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	No data available	
Water Solubility	Slightly soluble	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
1,2,4-Trichlorobenzene	4.05	
Vapor Pressure	2 hPa @ 50 °C	
Density / Specific Gravity	1.450	
Bulk Density	Not applicable	Liquid
Vapor Density	6.26 (Air = 1.0)	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

Molecular Formula	C6 H3 Cl3
Molecular Weight	181.45

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	None under normal processing.

10.4. Conditions to avoid

Incompatible products.

10.5. Incompatible materials

Strong oxidizing agents. Metals.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO₂). Phosgene. Hydrogen chloride gas.

SECTION 11: TOXICOLOGICAL INFORMATION

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11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
1,2,4-Trichlorobenzene	LD50 = 756 mg/kg (Rat)	LD50 = 6139 mg/kg (Rat)	-

(b) skin corrosion/irritation;

Category 2

(c) serious eye damage/irritation;

No data available

(d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

(e) germ cell mutagenicity;

No data available

(f) carcinogenicity;

No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

(g) reproductive toxicity;

No data available

(h) STOT-single exposure;

No data available

(i) STOT-repeated exposure;

No data available

Target Organs

No information available.

(j) aspiration hazard;

No data available

Symptoms / effects, both acute and delayed

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
1,2,4-Trichlorobenzene	LC50: 2.7 - 4.1 mg/L, 96h static	EC50: = 2.7 mg/L, 48h (Daphnia)	EC50: = 8.4 mg/L, 96h

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	(Lepomis macrochirus) LC50: = 6.57 mg/L, 96h static (Brachydanio rerio) LC50: = 4.8 mg/L, 96h (Oryzias latipes) LC50: 1.24 - 1.4 mg/L, 96h flow-through (Oncorhynchus mykiss) LC50: 1.67 - 4.34 mg/L, 96h flow-through (Pimephales promelas) LC50: 2.68 - 3.4 mg/L, 96h flow-through (Lepomis macrochirus) LC50: = 3.02 mg/L, 96h (Lepomis macrochirus) LC50: = 2.76 mg/L, 96h (Pimephales promelas) LC50: 3.4 - 4.77 mg/L, 96h static (Oncorhynchus mykiss)	magna)	(Desmodesmus subspicatus) EC50: = 1.4 mg/L, 96h (Pseudokirchneriella subcapitata) EC50: = 1.4 mg/L, 96h static (Pseudokirchneriella subcapitata) EC50: 11.1 - 36.2 mg/L, 72h (Pseudokirchneriella subcapitata)
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Component	Microtox	M-Factor
1,2,4-Trichlorobenzene	EC50 = 0.91 mg/L 24 h EC50 = 4.0 mg/L 30 min	

12.2. Persistence and degradability

Persistence

May persist, based on information available.

Degradation in sewage treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

May have some potential to bioaccumulate; Product has a high potential to bioconcentrate

Component	log Pow	Bioconcentration factor (BCF)
1,2,4-Trichlorobenzene	4.05	120 - 1320 dimensionless

12.4. Mobility in soil

. Is not likely mobile in the environment due its low water solubility. Is not likely mobile in the environment due its low water solubility and propensity to bind to soil particles

12.5. Results of PBT and vPvB assessment

No data available for assessment.

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but

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application specific.

Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number UN2321
14.2. UN proper shipping name TRICHLOROBENZENES, LIQUID
14.3. Transport hazard class(es) 6.1
14.4. Packing group III

ADR

14.1. UN number UN2321
14.2. UN proper shipping name TRICHLOROBENZENES, LIQUID
14.3. Transport hazard class(es) 6.1
14.4. Packing group III

IATA

14.1. UN number UN2321
14.2. UN proper shipping name TRICHLOROBENZENES, LIQUID
14.3. Transport hazard class(es) 6.1
14.4. Packing group III

14.5. Environmental hazards Dangerous for the environment
Product is a marine pollutant according to the criteria set by IMDG/IMO

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
1,2,4-Trichlorobenzene	120-82-1	204-428-0	-	-	X	X	KE-34063	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
1,2,4-Trichlorobenzene	120-82-1	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

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Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
1,2,4-Trichlorobenzene	120-82-1	-	Use restricted. See item 49. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	-

REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
1,2,4-Trichlorobenzene	120-82-1	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Component	ANNEX I - PART 1 List of chemicals subject to export notification procedure (referred to in Article 8)	ANNEX I - PART 2 List of chemicals qualifying for PIC notification (referred to in Article 11)	ANNEX I - PART 3 List of chemicals subject to the PIC procedure (referred to in Articles 13 and 14)
1,2,4-Trichlorobenzene 120-82-1 (>95)	sr — severe restriction i(2) — industrial chemical for public	-	-

<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32012R0649&qid=1604065742303>.

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

See table for values

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
1,2,4-Trichlorobenzene	WGK3	

Component	France - INRS (Tables of occupational diseases)
1,2,4-Trichlorobenzene	Tableaux des maladies professionnelles (TMP) - RG 9

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

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Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
1,2,4-Trichlorobenzene 120-82-1 (>95)	Prohibited and Restricted Substances Prohibited Halogenated Organic Compounds		

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed

H315 - Causes skin irritation

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Prepared By

Health, Safety and Environmental Department

Creation Date

24-Nov-2010

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28-Jan-2024

Revision Summary

New emergency telephone response service provider.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.

SAFETY DATA SHEET

1,2,4-Trichlorobenzene

Revision Date 28-Jan-2024

COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, Chemo (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer

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End of Safety Data Sheet